International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Practice Problems with Solutions

Chapter 6: Limits of sequences

Definitions and facts assumed

The solutions below are self-contained: every result used in a proof is stated where it is used. The following definitions and facts are assumed throughout and are not reproved.

Definitions.

(D1) A sequence $(a_n)_{n=m}^{\infty}$ of reals *converges to* a real L iff for every real $\varepsilon > 0$ there is an $N \geq m$ such that $|a_n - L| \leq \varepsilon$ for all $n \geq N$. We then write $L = \lim_{n\to\infty} a_n$, and call the sequence *convergent*; a sequence which converges to no real number is said to *diverge*.

(D2) A real number L is a *limit point* of $(a_n)_{n=m}^{\infty}$ iff for every real $\varepsilon > 0$ and every $N \geq m$ there is an $n \geq N$ with $|a_n - L| \leq \varepsilon$.

(D3) $(a_n)_{n=m}^{\infty}$ is *bounded* iff there is a real $M \geq 0$ with $|a_n| \leq M$ for all $n \geq m$.

(D4) The *extended reals* are $\mathbb{R}^* := \mathbb{R} \cup \{+\infty, -\infty\}$, ordered by the order of $\mathbb{R}$ together with $-\infty < x < +\infty$ for every real x, and $-\infty < +\infty$. We put $-(+\infty) := -\infty$ and $-(-\infty) := +\infty$.

(D5) For $E \subseteq \mathbb{R}^*$ the *supremum* $\sup(E)$ is defined thus. If $+\infty \in E$, then $\sup(E) := +\infty$. If $+\infty \notin E$ but $-\infty \in E$, then $\sup(E) := \sup(E \setminus \{-\infty\})$. If $E \subseteq \mathbb{R}$, then $\sup(E)$ is the least upper bound of E when E is non-empty and bounded above (this exists by the least upper bound property of $\mathbb{R}$); it is $+\infty$ when E is non-empty and not bounded above; and it is $-\infty$ when $E = \emptyset$. The *infimum* is $\inf(E) := -\sup(-E)$, where $-E := \{\, -x : x \in E \,\}$.

(D6) $(b_j)_{j=0}^{\infty}$ is a *subsequence* of $(a_n)_{n=0}^{\infty}$ iff there is a strictly increasing function $f : \mathbb{N} \to \mathbb{N}$ (that is, $f(j+1) > f(j)$ for all j) with $b_j = a_{f(j)}$ for all j.

(D7) *(Real exponents.)* Let $x > 0$ be real and let α be real. There is at least one sequence of rationals converging to α; for any such sequence (q_n) the limit $\lim_{n\to\infty} x^{q_n}$ exists, and its value is the same for every such sequence. We define

$$x^{\alpha} := \lim_{n\to\infty} x^{q_n}.$$

Facts.

(F1) $|a+b| \le |a| + |b|$, $|ab| = |a|\,|b|$, $|-a| = |a|$, and $|a| = 0$ if and only if $a = 0$.

(F2) *(Archimedean property.)* For every real x there is a positive integer N with $x \le N$.

(F3) *(Limit laws.)* If $a_n \to L$ and $b_n \to M$ then $a_n + b_n \to L + M$, $a_n b_n \to LM$, and $c\,a_n \to cL$ for every real c; and if moreover $M \neq 0$ and every $b_n \neq 0$, then $a_n / b_n \to L/M$.

(F4) Limits preserve non-strict inequalities: if $a_n \to L$ and $a_n \ge c$ for all n, then $L \ge c$; and likewise with $\le$.

(F5) Every finite sequence of reals is bounded.

(F6) Every non-empty set of natural numbers has a least element.

(F7) *(Bernoulli's inequality.)* $(1+h)^n \ge 1 + nh$ for every real $h \ge 0$ and every integer $n \ge 1$.

(F8) For a real $x \ge 0$ and an integer $n \ge 1$, the number $x^{1/n}$ is the unique non-negative real whose nth power is x. Moreover $x > y \ge 0$ if and only if $x^{1/n} > y^{1/n}$, and $(xy)^{1/n} = x^{1/n} y^{1/n}$.

(F9) For a real $x > 0$ and rationals q, r: the number x^q is a positive real, and

$$x^0 = 1, \qquad 1^q = 1, \qquad x^{q+r} = x^q x^r,$$

$$x^{a/b} = (x^{1/b})^a \quad \text{for integers } a \text{ and } b \ge 1.$$

Moreover, if $x > 1$ then $x^q > x^r$ exactly when $q > r$.

Part A consists of the results proved in lectures, set as exercises. Part B contains further questions built on them.

Part A. The results of the chapter

Question (1). Let $(a_n)_{n=m}^{\infty}$ be a real sequence starting at some integer index m, and let $L \neq L'$ be two distinct real numbers. Show that it is not possible for $(a_n)_{n=m}^{\infty}$ to converge to L while also converging to L'.

Suppose, for contradiction, that $a_n \to L$ and $a_n \to L'$ with $L \neq L'$. Then $|L - L'| > 0$ by (F1), so

$$\varepsilon := \frac{|L - L'|}{3}$$

is a positive real number.

By (D1) applied to each limit there are $N_1, N_2 \geq m$ with $|a_n - L| \leq \varepsilon$ for all $n \geq N_1$, and $|a_n - L'| \leq \varepsilon$ for all $n \geq N_2$. Choose any $n \geq \max(N_1, N_2)$; such an n exists since the sequence is infinite. By (F1),

$$|L - L'| \;=\; |(L - a_n) + (a_n - L')| \;\leq\; |L - a_n| + |a_n - L'| \;\leq\; 2\varepsilon \;=\; \tfrac{2}{3}\,|L - L'|,$$

using $|L - a_n| = |a_n - L|$. Subtracting $\frac{2}{3}|L - L'|$ gives $\frac{1}{3}|L - L'| \leq 0$, hence $|L - L'| \leq 0$. But $|L - L'| \geq 0$, so $|L - L'| = 0$ and therefore $L = L'$ by (F1) — contradicting $L \neq L'$.

Question (2). Let E be a subset of $\mathbb{R}^*$. Show that the following statements are true.

(a) For every $x \in E$ we have $x \leq \sup(E)$ and $x \geq \inf(E)$.

(b) Suppose that $M \in \mathbb{R}^*$ is an upper bound for E, that is, $x \leq M$ for all $x \in E$. Then we have $\sup(E) \leq M$.

(c) Suppose that $M \in \mathbb{R}^*$ is a lower bound for E, that is, $x \geq M$ for all $x \in E$. Then we have $\inf(E) \geq M$.

We first prove the two statements about sup, then deduce those about inf from the definition $\inf(E) = -\sup(-E)$ in (D5).

(a), supremum half. Let $x \in E$; we show $x \leq \sup(E)$.

If $x = -\infty$ there is nothing to prove, since $-\infty$ is the least element of $\mathbb{R}^*$ by (D4). If $x = +\infty$, then $+\infty \in E$, so $\sup(E) = +\infty$ by (D5) and $x \leq \sup(E)$.

So suppose x is real. If $+\infty \in E$ then $\sup(E) = +\infty \geq x$. Otherwise set $E' := E \setminus \{-\infty\}$, so that $E' \subseteq \mathbb{R}$ and $x \in E'$; and by (D5), $\sup(E) = \sup(E')$ whether or not $-\infty \in E$. Now E' is non-empty, as $x \in E'$. If E' is bounded above, then $\sup(E')$ is an upper bound for E' and hence $x \leq \sup(E')$; if E' is not bounded above then $\sup(E') = +\infty \geq x$. Either way $x \leq \sup(E)$.

(b). Let $M \in \mathbb{R}^*$ be an upper bound for E.

Case $+\infty \in E$. Then $+\infty \leq M$, which by (D4) forces $M = +\infty$; and $\sup(E) = +\infty \leq M$.

Case $+\infty \notin E$. Put $E' := E \setminus \{-\infty\} \subseteq \mathbb{R}$, so $\sup(E) = \sup(E')$ by (D5), and M is an upper bound for E' as well.

If $E' = \emptyset$ then $\sup(E') = -\infty \leq M$.

If $E' \neq \emptyset$, pick $y \in E' \subseteq \mathbb{R}$. Since $y \leq M$ and y is real, $M \neq -\infty$. If $M = +\infty$ then $\sup(E') \leq +\infty = M$. If M is real, then M is a real upper bound for the non-empty set $E' \subseteq \mathbb{R}$, so E' is bounded above and $\sup(E')$ is its *least* upper bound; hence $\sup(E') \leq M$.

In all cases $\sup(E) \leq M$.

(c), and the infimum half of (a). Observe first that for $u, v \in \mathbb{R}^*$ we have $u \leq v$ if and only if $-v \leq -u$; this is immediate from (D4), since negation reverses the order of $\mathbb{R}$ and exchanges $+\infty$ with $-\infty$.

Let $x \in E$. Then $-x \in -E$, so $-x \leq \sup(-E)$ by the part of (a) already proved. Negating reverses this: $-\sup(-E) \leq x$, that is, $\inf(E) \leq x$. This is the infimum half of (a).

Now let M be a lower bound for E, so $M \leq x$ for every $x \in E$. Negating, $-x \leq -M$ for every $x \in E$, that is, $-M$ is an upper bound for $-E$. By (b), $\sup(-E) \leq -M$. Negating again gives $M \leq -\sup(-E) = \inf(E)$, which is (c).

Question (3). Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, and let $(b_k)_{k=p}^{\infty}$ be another sequence of real numbers such that each b_k is a limit point of $(a_n)_{n=m}^{\infty}$. Let c be a limit point of $(b_k)_{k=p}^{\infty}$. Prove that c is also a limit point of $(a_n)_{n=m}^{\infty}$. (In other words, limit points of limit points are themselves limit points of the original sequence.)

By (D2) we must show: for every $\varepsilon > 0$ and every $N \geq m$ there is an $n \geq N$ with $|a_n - c| \leq \varepsilon$. So let $\varepsilon > 0$ and $N \geq m$ be given.

Step 1: choose a b near c. Since c is a limit point of $(b_k)_{k=p}^{\infty}$, applying (D2) with

tolerance $\varepsilon/2$ and threshold p gives an index $k \geq p$ with

$$|b_k - c| \;\leq\; \frac{\varepsilon}{2}.$$

Step 2: choose an a near that b. Fix the index k just found. By hypothesis b_k is a limit point of $(a_n)_{n=m}^{\infty}$, so applying (D2) with tolerance $\varepsilon/2$ and threshold N gives an index $n \geq N$ with

$$|a_n - b_k| \;\leq\; \frac{\varepsilon}{2}.$$

Step 3: combine. By (F1),

$$|a_n - c| \;\leq\; |a_n - b_k| + |b_k - c| \;\leq\; \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \;=\; \varepsilon,$$

and $n \geq N$. Since ε and N were arbitrary, c is a limit point of $(a_n)_{n=m}^{\infty}$.

Remark. The order of the two steps matters. The index k must be chosen first, using the threshold p that comes with the sequence (b_k); only then can n be chosen using the threshold N handed to us. Choosing n first would leave no way to control $|a_n - b_k|$ for a k produced afterwards.

Question (4). Show that $\lim_{n\to\infty} 1/n^{1/k} = 0$ for every integer $k \geq 1$.

We first note a non-strict form of the monotonicity in (F8): if $s \geq t \geq 0$ then $s^{1/k} \geq t^{1/k}$. Indeed, if instead $s^{1/k} < t^{1/k}$, then (F8) applied to the pair t and s would give $t > s$, contrary to hypothesis.

Now let $k \geq 1$ and let $\varepsilon > 0$ be real. Put $t := 1/\varepsilon$, a positive real, so that $t^k > 0$. By (F2) there is a positive integer N with $t^k \leq N$.

Let $n \geq N$. Then $n \geq t^k \geq 0$, so by the monotonicity just stated,

$$n^{1/k} \;\geq\; \left(t^k\right)^{1/k} \;=\; t,$$

the last equality because t is a non-negative real whose kth power is t^k, and (F8) says $(t^k)^{1/k}$ is the only such number. Since $t = 1/\varepsilon > 0$ and $n^{1/k} \geq t > 0$, taking reciprocals reverses the inequality:

$$\left|\frac{1}{n^{1/k}} - 0\right| \;=\; \frac{1}{n^{1/k}} \;\leq\; \frac{1}{t} \;=\; \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, (D1) gives $\lim_{n\to\infty} 1/n^{1/k} = 0$.

Question (5). Show that for any $x > 0$ we have $\lim_{n\to\infty} x^{1/n} = 1$.

Case $x = 1$. By (F8) the number $1^{1/n}$ is the unique non-negative real whose nth power is 1; since $1^n = 1$, that number is 1. So the sequence is constantly 1 and converges to 1.

Case $x > 1$. For $n \geq 1$ put $y_n := x^{1/n} - 1$. Since $x > 1 = 1^{1/n}$, (F8) gives $x^{1/n} > 1$, so $y_n > 0$. Also $(1 + y_n)^n = (x^{1/n})^n = x$ by (F8). By (F7),

$$x \;=\; (1 + y_n)^n \;\geq\; 1 + n y_n,$$

so that

$$0 \;<\; y_n \;\leq\; \frac{x-1}{n}.$$

Let $\varepsilon > 0$. By (F2) choose a positive integer N with $(x-1)/\varepsilon \leq N$. For $n \geq N$,

$$|x^{1/n} - 1| \;=\; y_n \;\leq\; \frac{x-1}{n} \;\leq\; \frac{x-1}{N} \;\leq\; \varepsilon.$$

So $\lim_{n\to\infty} x^{1/n} = 1$.

Case $0 < x < 1$. Put $z := 1/x > 1$. By (F8),

$$x^{1/n} z^{1/n} \;=\; (xz)^{1/n} \;=\; 1^{1/n} \;=\; 1,$$

so $x^{1/n} = 1/z^{1/n}$. By the previous case $z^{1/n} \to 1$, and $1 \neq 0$, and every $z^{1/n} > 0$ by (F8); so the quotient law in (F3), applied to the constant sequence 1 divided by $(z^{1/n})$, gives

$$\lim_{n\to\infty} x^{1/n} \;=\; \frac{1}{1} \;=\; 1.$$

Question (6). Show that $\lim_{n\to\infty} 1/n^q = 0$ for any rational $q > 0$. Conclude that the sequence $(n^q)_{n=1}^{\infty}$ does not converge to any real number.

The limit of $1/n^q$. Write $q = a/b$ with a and b positive integers, possible since $q > 0$. By (F9), $n^q = (n^{1/b})^a$ for every $n \geq 1$, and $n^q > 0$, so

$$\frac{1}{n^q} \;=\; \Big(\frac{1}{n^{1/b}}\Big)^a.$$

By Question (4), $\lim_{n\to\infty} 1/n^{1/b} = 0$. Applying the product law in (F3) repeatedly — $a - 1$ times, to the sequence $(1/n^{1/b})$ multiplied by itself — gives $\lim_{n\to\infty}(1/n^{1/b})^a = 0^a = 0$. Hence $\lim_{n\to\infty} 1/n^q = 0$.

(n^q) **does not converge.** Suppose $n^q \to L$ for some real L.

First, $n^q \geq 1$ for every $n \geq 1$: for $n = 1$ we have $1^q = 1$ by (F9), and for $n \geq 2 > 1$ the last clause of (F9) with $r := 0$ gives $n^q > n^0 = 1$, as $q > 0$.

By (F4) it follows that $L \geq 1$, so in particular $L \neq 0$. Since every $n^q > 0$, the quotient law in (F3) applies to the constant sequence 1 divided by (n^q) and gives

$$\lim_{n\to\infty} \frac{1}{n^q} \;=\; \frac{1}{L}.$$

But the first part showed this limit is 0, so by Question (1) — uniqueness of limits — we get $1/L = 0$. Multiplying by L gives $1 = 0$, which is false. Hence no such L exists.

Question (7). Can you find two sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ which are not the same sequence, but such that each is a subsequence of the other?

Yes. Define

$$a_n := \begin{cases} 0, & n \text{ even}, \\ 1, & n \text{ odd}, \end{cases} \qquad b_n := \begin{cases} 1, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases}$$

so that $(a_n) = 0, 1, 0, 1, \ldots$ and $(b_n) = 1, 0, 1, 0, \ldots$.

They are not the same sequence. $a_0 = 0$ while $b_0 = 1$, and $0 \neq 1$.

(b_n) *is a subsequence of* (a_n). Take $f(n) := n + 1$, which maps $\mathbb{N}$ to $\mathbb{N}$ and is strictly increasing. If n is even then $n + 1$ is odd, so $a_{f(n)} = a_{n+1} = 1 = b_n$; if n is odd then $n + 1$ is even, so $a_{f(n)} = 0 = b_n$. So $b_n = a_{f(n)}$ for every n, which is (D6).

(a_n) *is a subsequence of* (b_n). The same $f(n) = n + 1$ works: if n is even then $b_{n+1} = 0 = a_n$, and if n is odd then $b_{n+1} = 1 = a_n$.

Remark. So "is a subsequence of" is not antisymmetric, and is therefore not a partial order on sequences. What is true is that two sequences each a subsequence of the other have the same limit points.

Question (8). Let $(a_n)_{n=0}^{\infty}$ be a sequence which is not bounded. Show that there exists a subsequence $(b_n)_{n=0}^{\infty}$ of $(a_n)_{n=0}^{\infty}$ such that every b_n is non-zero and $\lim_{n\to\infty} 1/b_n$ exists and is equal to zero. (Hint: for each natural number j, recursively introduce the quantity $n_j := \min\{n \in \mathbb{N} : |a_n| \geq j + 1;\ n > n_{j-1}\}$ (omitting the condition $n > n_{j-1}$ when $j = 0$), first explaining why the set $\{n \in \mathbb{N} : |a_n| \geq j + 1;\ n > n_{j-1}\}$

is non-empty. Then set $b_j := a_{n_j}$.)

The sets are non-empty. Following the hint, let $j \in \mathbb{N}$ and suppose $n_0 < n_1 < \cdots < n_{j-1}$ have already been chosen. Put

$$S_j := \{\, n \in \mathbb{N} : |a_n| \geq j+1 \text{ and } n > n_{j-1} \,\},$$

omitting the second condition when $j = 0$.

Suppose S_j were empty. If $j = 0$, this says $|a_n| < 1$ for every n, so (a_n) would be bounded by (D3), contrary to hypothesis. If $j \geq 1$, it says $|a_n| < j+1$ for every $n > n_{j-1}$. The remaining terms $a_0, a_1, \ldots, a_{n_{j-1}}$ form a finite sequence, hence are bounded by (F5): there is $M \geq 0$ with $|a_n| \leq M$ for $n \leq n_{j-1}$. Then $|a_n| \leq \max(M, j+1)$ for every n, so again (a_n) would be bounded, contrary to hypothesis. So $S_j \neq \emptyset$.

The construction. By (F6) the non-empty set $S_j \subseteq \mathbb{N}$ has a least element; put $n_j := \min S_j$. By construction $n_j > n_{j-1}$ for $j \geq 1$, so $j \mapsto n_j$ is strictly increasing, and

$$|a_{n_j}| \;\geq\; j+1 \qquad \text{for every } j \in \mathbb{N}.$$

Setting $b_j := a_{n_j}$ therefore gives a subsequence of (a_n) in the sense of (D6).

The limit. For every $j \in \mathbb{N}$ we have $|b_j| \geq j+1 \geq 1 > 0$, so $b_j \neq 0$ and $1/b_j$ is defined. Moreover, by (F1),

$$\left|\frac{1}{b_j} - 0\right| \;=\; \frac{1}{|b_j|} \;\leq\; \frac{1}{j+1}.$$

Let $\varepsilon > 0$. By (F2) choose a positive integer J with $1/\varepsilon \leq J$, so that $1/J \leq \varepsilon$. For $j \geq J$ we get $|1/b_j - 0| \leq 1/(j+1) \leq 1/J \leq \varepsilon$. Hence $\lim_{j\to\infty} 1/b_j = 0$ by (D1).

Remark. The clause $n > n_{j-1}$ in the hint is what forces the indices to increase. Without it one could pick the same index again and again, and the result would not be a subsequence. The bound $|a_n| \geq j+1$, rather than $|a_n| \geq j$, is what makes every b_j non-zero: with $|a_n| \geq 0$ at $j = 0$ the first term b_0 could be 0, and then $1/b_0$ would not be defined.

Part B. Further questions

Question (9). Let $(a_n)_{n=0}^{\infty}$ be a sequence of reals.

(a) Show that if (a_n) converges to L, then every subsequence of (a_n) also converges to L.

(b) Deduce the following divergence test: if (a_n) has two subsequences converging to different limits, then (a_n) diverges.

(c) Use (b) to show that the sequence $a_n := (-1)^n$ diverges.

(a) Let $(a_{n_j})_{j=0}^{\infty}$ be a subsequence, so that $j \mapsto n_j$ is strictly increasing by (D6).

We first check that $n_j \geq j$ for every j. This holds at $j = 0$, since n_0 is a natural number and so $n_0 \geq 0$. And if $n_j \geq j$, then $n_{j+1} > n_j \geq j$; as these are natural numbers, $n_{j+1} > j$ forces $n_{j+1} \geq j + 1$. The claim follows by induction.

Now let $\varepsilon > 0$. By (D1) there is N with $|a_n - L| \leq \varepsilon$ for all $n \geq N$. For $j \geq N$ we have $n_j \geq j \geq N$, hence $|a_{n_j} - L| \leq \varepsilon$. So $a_{n_j} \to L$.

(b) Suppose (a_n) converged, to L say. By (a) every subsequence would converge to L; so two subsequences with limits $L_1 \neq L_2$ would give $L_1 = L = L_2$, contradicting $L_1 \neq L_2$. (That a sequence cannot have two distinct limits is Question (1).)

(c) The subsequence of even indices, $f(j) := 2j$, is constantly 1 and so converges to 1; the subsequence of odd indices, $f(j) := 2j + 1$, is constantly -1 and so converges to -1. Both index maps are strictly increasing, so these are subsequences in the sense of (D6). Since $1 \neq -1$, part (b) gives divergence.

Remark. This is the standard way of showing that a sequence diverges: exhibit two subsequences with different limits. It is usually quicker than arguing directly from (D1), which requires ruling out every candidate limit L at once.

Question (10). Let $E, F \subseteq \mathbb{R}^*$.

(a) Show that if $E \subseteq F$ then $\sup(E) \leq \sup(F)$ and $\inf(E) \geq \inf(F)$.

(b) Show that if E is non-empty then $\inf(E) \leq \sup(E)$.

(c) Show that if E is non-empty and $\sup(E) = \inf(E) = c$ for some $c \in \mathbb{R}^*$, then $E = \{c\}$.

(d) Show that (b) fails for $E = \emptyset$.

(a) Let $x \in E$. Then $x \in F$, so $x \leq \sup(F)$ by Question (2)(a). As $x \in E$ was arbitrary, $\sup(F)$ is an upper bound for E, and Question (2)(b) gives $\sup(E) \leq \sup(F)$.

Likewise every $x \in E$ satisfies $x \geq \inf(F)$ by Question (2)(a), so $\inf(F)$ is a lower bound for E, and Question (2)(c) gives $\inf(E) \geq \inf(F)$.

(b) Since $E \neq \emptyset$, choose $x \in E$. By Question (2)(a), $\inf(E) \leq x \leq \sup(E)$.

(c) Let $x \in E$. By Question (2)(a), $c = \inf(E) \leq x \leq \sup(E) = c$, so $x = c$. Hence $E \subseteq \{c\}$; and E is non-empty, so $E = \{c\}$.

(d) For $E = \emptyset$ we have $\sup(E) = -\infty$ by (D5). Also $-E = \emptyset$, so $\inf(E) = -\sup(\emptyset) = -(-\infty) = +\infty$. Thus $\inf(E) = +\infty > -\infty = \sup(E)$, and the inequality of (b) fails.

Remark. Part (d) explains the non-emptiness hypothesis in (b) and (c). It is not a defect of the definition: $\sup(\emptyset) = -\infty$ is exactly what makes (a) true without exception.

Question (11). Let $x > 0$ be real, and recall from (D7) that for a real number α we set $x^\alpha := \lim_{n\to\infty} x^{q_n}$, where (q_n) is any sequence of rationals converging to α. Show that:

(a) $x^\alpha > 0$;

(b) $x^{\alpha+\beta} = x^\alpha x^\beta$ for all reals α, β;

(c) if $x > 1$ and $\alpha > 0$ then $x^\alpha \geq 1$.

By (D7) the value of x^α is the same whichever rational sequence we use, so below we pick a convenient one for each exponent.

(b) first. Choose rationals $q_n \to \alpha$ and $r_n \to \beta$. By the sum law in (F3), $q_n + r_n \to \alpha + \beta$, and each $q_n + r_n$ is rational. So $(q_n + r_n)$ is an admissible sequence for $\alpha + \beta$, and

$$x^{\alpha+\beta} \;=\; \lim_{n\to\infty} x^{q_n+r_n} \;=\; \lim_{n\to\infty}\left(x^{q_n}x^{r_n}\right) \;=\; \left(\lim_{n\to\infty} x^{q_n}\right)\left(\lim_{n\to\infty} x^{r_n}\right) \;=\; x^\alpha x^\beta,$$

where the second equality is (F9) and the third is the product law in (F3).

(a) Each $x^{q_n} > 0$ by (F9), so $x^\alpha \geq 0$ by (F4). Now apply (b) with $\beta := -\alpha$: the constant sequence $0, 0, 0, \ldots$ is a sequence of rationals converging to 0, so $x^0 = \lim_{n\to\infty} x^0 = 1$ by (D7) and (F9), the inner x^0 being the rational power. Hence $x^\alpha x^{-\alpha} = 1$, so $x^\alpha \neq 0$ and therefore $x^\alpha > 0$.

(c) Let $x > 1$ and $\alpha > 0$, and pick rationals $q_n \to \alpha$. Applying (D1) with

$\varepsilon := \alpha/2 > 0$, there is N with $|q_n - \alpha| \le \alpha/2$ for $n \ge N$, hence $q_n \ge \alpha/2 > 0$ for such n. By the last clause of (F9) with $r := 0$, $x^{q_n} > x^0 = 1$ for $n \ge N$.

Since discarding finitely many terms changes neither convergence nor limit, (F4) applied to $(x^{q_n})_{n \ge N}$ gives $x^\alpha \ge 1$.

Remark. Only ≥ 1 follows here, since (F4) concerns non-strict inequalities: $0 < 1/n$ for every $n \ge 1$, yet both sides tend to 0. In fact $x^\alpha > 1$; compare x^α with x^r for a rational r with $0 < r < \alpha$.